Computational Neuroscience — Week 3

Neural Decoding

Reading the Stimulus Back Out of the Spikes

These notes cover lectures 1–3 — deciding between two alternatives from a single noisy neuron, decoding a continuous stimulus from a whole population, and reconstructing a stimulus that varies in time — followed by Fred Rieke's guest lecture on single-photon detection. Green boxes give the plain-words version of each idea before the mathematics; rust boxes flag the places where the obvious reading is the wrong one, including one slip on the slides themselves.

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Part I

Deciding Between Two Alternatives

Lecture 1 — signal detection theory, the likelihood ratio, accumulating evidence to a bound, priors, and costs.

1. The Question

Week 2 asked how a stimulus is turned into spikes. This week runs the arrow backwards.

Decoding

Given the observed neural response, how well can we recover what the stimulus was?

Note who is doing the recovering. There are two readers, and they are easy to conflate:

- **The experimenter**, sitting outside the animal with a spike train on a screen, asking *how much information is in here?*
- **The animal**, which has nothing *but* the spikes and must actually commit to an action.

The mathematics is the same for both. What makes this lecture interesting is that the two answers turn out to be numerically almost identical — a single neuron in the right part of cortex does about as well as the whole monkey.

The lecture builds the decoder in four steps, each one adding something the previous step ignored:

1. One observation, two choices, nothing known in advance → *signal detection theory*
2. The best possible way to use that one observation → *the likelihood ratio*
3. Many observations, and time to wait → *accumulating evidence to a bound*
4. Some outcomes are rarer, and some mistakes hurt more → *priors and costs*

2. Do I Stay or Do I Go?

You are walking in the park and hear a rustle in the bushes. It could be the breeze. It could be a tiger. You have to choose.

[Lecture slide omitted from the public copy.] The problem, as the lecture sets it up. A rustle in the bushes: breeze, or tiger? Overlaid are the two distributions of possible sounds along an axis s. They overlap, and that overlap is the whole difficulty — a quiet tiger and a loud breeze make the same sound.

The idea in plain words

If breeze-rustles and tiger-rustles sounded completely different there would be nothing to discuss: hear the sound, know the cause. The difficulty is that a quiet tiger and a loud breeze produce the *same* sound. No amount of cleverness recovers the truth with certainty from a single sound. The best you can do is be wrong as rarely as possible — and, later, be wrong in the cheaper direction.

Now make it harder in the way biology actually makes it harder: you do not have direct access to the sound. All you have is your neurons.

3. The Experiment: Random Dots and Area MT

The classic study is **Britten et al. (1992)**, designed to probe how noisy sensory information is represented by noisy sensory neurons — and how the animal's decision relates to that representation.

The task

A monkey fixates the centre of a screen and watches a field of moving dots. It is trained to **saccade** (flick its eyes) in the direction the dot pattern is moving, and is rewarded when the direction is right.

The experimenters control the difficulty with **coherence** — the percentage of dots moving together in the chosen direction. The rest move randomly.

[Lecture slide omitted from the public copy.] Coherence sets the noise level. At 100% every dot moves together and the answer is obvious. At 0% the dots move at random and there is genuinely no correct answer. The interesting regime is in between. Britten et al. '92.

Watch out — 0% coherence is not a hard trial — it is an empty one

At zero coherence there is no motion signal at all, so the monkey is guessing and scores 50%. This matters when reading the performance curve later: the curve starts at 0.5, not at 0. A chance-level score is the floor of the plot, not a failure.

The recording

Spikes were recorded from **area MT** (also called V5), a region of monkey visual cortex highly sensitive to visual motion. The task was repeated many times, and on each trial the number of spikes the neuron fired was counted.

[Lecture slide omitted from the public copy.] Spike-count histograms at three coherences. Filled bars are trials where the monkey chose one direction, open bars the other. At 12.8% the two are clearly displaced; by 0.8% they sit almost on top of one another. Less motion signal, more overlap, harder decision. Britten et al. '92.

The idea in plain words

This is the tiger and the breeze again, now in the currency the brain actually uses. Instead of "how loud was the rustle" the axis is "how many spikes did this neuron fire". Upward motion tends to produce more spikes, downward motion fewer, but any particular spike count is consistent with both.
As coherence falls the two histograms slide together. Near zero coherence they sit almost on top of one another: the visual signal that distinguishes the two directions is tiny, so the firing rates are nearly the same.

The result that makes the lecture

[Lecture slide omitted from the public copy.] Fraction correct against coherence. Filled circles: the monkey's behaviour. Open circles: an ideal observer decoding one MT neuron. The two curves lie almost on top of each other. Britten et al. '92.

A single neuron, decoded well, is about as good as the entire animal. That is the finding the rest of the lecture unpacks — and it immediately raises a question the lecture flags and postpones.

But getting from a pair of overlapping histograms to a single number for "fraction correct" needs a rule for turning a firing rate into a choice. That rule is the decoding step.

4. Signal Detection Theory

Strip the data down to a cartoon. Let r be the observed response (the firing rate on this trial), and write the two **conditional distributions**:

$$p(r \mid +) \quad \text{and} \quad p(r \mid -)$$

read as "the probability of seeing response r *given that* the stimulus was upward-moving", and likewise for downward. The vertical bar means "given that". Their averages are $\langle r \rangle_+$ and $\langle r \rangle_-$.

[Lecture slide omitted from the public copy.] The two response distributions, $p(r \mid -)$ in blue and $p(r \mid +)$ in red, with their means marked. Any single observed rate r is consistent with both.

Where to put the threshold

A decoding rule here is just a cut point. Pick a value z; if $r \geq z$ answer $+$, otherwise answer $-$. The only question is where to put it.

Every choice of z produces two kinds of error, and they trade off against each other:

Name	What happened	Its size
Correct detection	truth $+$, you said $+$	$P(r \geq z \mid +)$
False alarm	truth $-$, you said $+$	$P(r \geq z \mid -)$

Both are areas under a curve to the right of z — one under the red curve, one under the blue. Slide z to the left and you catch more real $+$ trials but also raise more false alarms. Slide it right and both fall. You cannot reduce one without raising the other; that is forced by the overlap.

Writing $b(z) = P(r \geq z \mid +)$ and $a(z) = P(r \geq z \mid -)$, the total probability of being right is

$$P[\text{correct}] \;=\; P[+]\, b(z) \;+\; P[-]\,\big(1 - a(z)\big).$$

The idea in plain words

Read it as two ways of being right, weighted by how often each situation arises. Either the stimulus really was $+$ (which happens a fraction $P[+]$ of the time) and you correctly said $+$; or it really was $-$ (a fraction $P[-]$ of the time) and you did *not* raise a false alarm — which is what $1 - a(z)$ counts.

Maximising this expression puts z at the point where the two curves cross. For the moment take that on trust; the next section shows exactly why, and the answer turns out to be more general than "the crossing point".

5. The Likelihood Ratio

The conditional probabilities $p(r \mid +)$ and $p(r \mid -)$ have a name.

Likelihood

How likely we were to observe the data r — our measured firing rate — given a particular cause of the stimulus.

Watch out — a likelihood is not a probability of the stimulus

$p(r \mid +)$ is the probability of *the response*, assuming the stimulus was $+$. It is emphatically not the probability that the stimulus was $+$. That other quantity, $P(+ \mid r)$, is what you actually want, and getting to it requires Bayes' rule and a prior — which is exactly the gap the last two sections of this lecture close.

Walking along the axis

Now look at the two curves again and walk from left to right.

Far to the left, $p(r \mid -)$ is the larger of the two, so any r down there is better explained by a $-$ stimulus: answer $-$. Keep walking. At some point the curves cross, and past it $p(r \mid +)$ is larger: answer $+$.

So the rule "choose the stimulus whose likelihood is larger" and the rule "put a threshold at the crossing point" are *the same rule*. That is why z goes where it does.

Rewriting it as one number

Rather than comparing two curves, take their ratio:

$$\ell(r) = \frac{p(r \mid +)}{p(r \mid -)} \qquad \text{answer } + \text{ when } \ell(r) > 1.$$

The idea in plain words

The ratio collapses two curves into one running score. It asks: *how many times better does the + hypothesis explain what I just saw than the − hypothesis does?* Bigger than 1 means + explains it better. Equal to 1 is the crossing point — the two hypotheses explain the observation equally well and you have learned nothing.
Nothing has changed mathematically; comparing $p(r \mid +)$ with $p(r \mid -)$ and comparing their ratio with 1 are the same comparison. What is gained is a *single number* whose threshold can now be moved away from 1 as more considerations arrive. Both remaining sections do exactly that.

[Lecture slide omitted from the public copy.] The likelihood ratio test, with the shaded false-alarm region and the threshold z. Written in by hand: answer + when $p(r \mid +)/p(r \mid -) > 1$. The Neyman–Pearson lemma guarantees no other test does better.

Neyman–Pearson lemma

Of all possible ways of deciding between two simple hypotheses, the likelihood ratio test is the most efficient: it has the most **power** for a given **size**.

Watch out — the slide's definition of *power* is the wrong way round

The deck glosses power as "the probability of a false negative". Power is the probability of *avoiding* a false negative — of correctly detecting a real signal, $P(r \geq z \mid +)$. If β is the false-negative rate then power is $1 - \beta$.
Size on the slide is right: the false-positive rate, $P(r \geq z \mid -)$, conventionally α.
So the lemma says: fix how many false alarms you are willing to tolerate, and no test catches more real signals than this one. Read with the slide's wording, the lemma would be claiming the test *maximises* the miss rate, which is backwards.

This closes the loop on the experiment. Apply the likelihood ratio test to the MT spike-count histograms, sweep the threshold, and you get the open circles on the performance plot — the "neuronal" curve that tracks the monkey.

6. Accumulating Evidence Over Time

Everything so far assumed a single snapshot and an immediate answer. Drop that assumption: suppose you can keep listening.

[Lecture slide omitted from the public copy.] Accumulating evidence, drawn during the lecture. Top left: the two overlapping distributions and the log likelihood ratio for one observation. The trace below wanders

between a tiger bound (red, above) and a breeze bound (blue, below), and the decision is made when it first touches one.

Why the logarithm

Each new observation s gives its own likelihood ratio $\ell(s) = p(s \mid \text{tiger})/p(s \mid \text{breeze})$. If successive samples are independent, the evidence from a whole sequence is the ratios *multiplied* together.

Multiplying many numbers is awkward, so take logarithms, which turn products into sums:

$$\log \prod_i \ell(s_i) \;=\; \sum_i \log \ell(s_i).$$

The idea in plain words

The log turns evidence into something that simply adds up, like a running tally. Start at zero, meaning no evidence either way.

- An observation favouring the tiger has $\ell > 1$, so $\log \ell > 0$ — the tally ticks **up**.
- An observation favouring the breeze has $\ell < 1$, so $\log \ell < 0$ — the tally ticks **down**.
- An uninformative observation has $\ell = 1$ and $\log \ell = 0$ — the tally does not move, which is exactly right.

This is the one place the logarithm earns its keep: it is what makes "accumulating evidence" literal addition rather than repeated multiplication.

The random walk

The tally wanders. A growl pushes it up, a rustle of leaves pushes it back down. It is a random walk with a drift, and the drift is set by which hypothesis is actually true. Put a bound on each side and stop the moment one is touched.

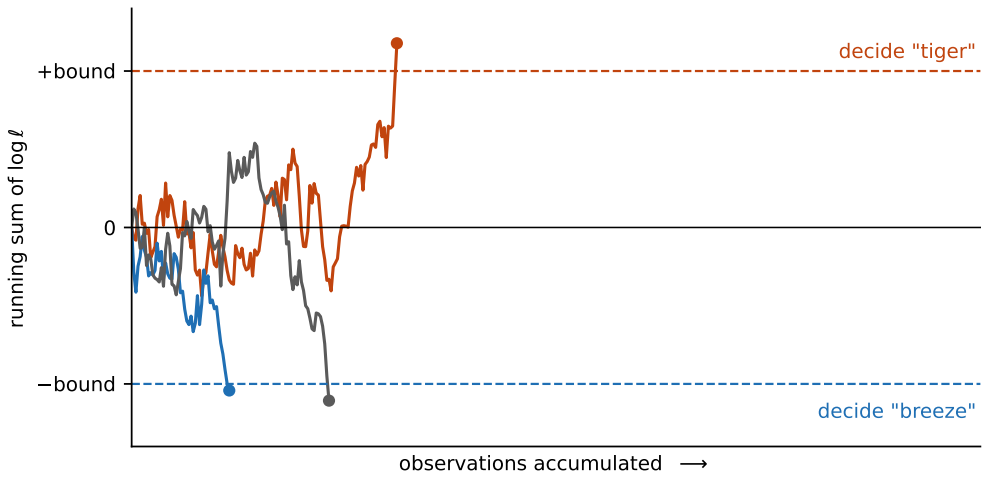


Figure 1: *Three simulated trials. The tally starts at zero and wanders; the walk stops when it first touches a bound. The rust trace has a slight upward drift and ends at "tiger"; the blue trace drifts down and ends quickly at "breeze"; the grey trace has almost no drift, wanders a long time, and only eventually commits. Weak evidence means slow decisions.*

The idea in plain words

The bound *is* your confidence requirement, and it buys accuracy with time. Set the bounds far apart and you demand a great deal of evidence: you will rarely be wrong, but you will be slow. Set them close and you decide fast and sloppily. This single knob produces the speed–accuracy trade-off that turns up everywhere in psychology.

Evidence that the brain does this

Kiani, Hanks & Shadlen (2006) ran nearly the same task with one change: the monkey could respond *whenever it wanted*. It was under time pressure — reward comes on answering, and dawdling triggers a time-out with no juice.

Recordings were made not from MT but from **LIP** (lateral intraparietal cortex), part of the circuitry for planning and executing eye movements. For each neuron the experimenters located the patch of space it responded to, and placed the saccade target there — so the neuron is tied to *the specific choice* the monkey will report.

[Lecture slide omitted from the public copy.] LIP firing rates (Kiani, Hanks & Shadlen 2006). Left: the reaction-time task. Right: rate against time, one colour per coherence. Aligned to the dots appearing the traces ramp at different slopes; aligned to the saccade they converge on a common peak.

Two features of the firing rates matter, and they are the two halves of the random-walk picture:

- **Aligned to when the dots appear**, the rate ramps up gradually, and the *slope* depends on coherence. Strong motion, steep ramp; weak motion, shallow ramp. The firing rate behaves like the running tally, filling up faster when evidence is stronger.
- **Aligned to the saccade**, the traces converge and peak at a *common* value regardless of coherence. Whatever the difficulty, the monkey moves when the rate reaches the same level.

The idea in plain words

That common peak is the striking part. It is what a bound looks like in spikes: the rate climbs until it hits a fixed threshold of confidence, and at that point the monkey commits. Easy trials reach it quickly, hard trials slowly, but everyone stops at the same line.

7. Priors: Building In What We Already Know

Back to one observation — and to a problem that has been sitting in plain sight. The rule $\ell(r) > 1$ treats tiger and breeze as equally plausible before you hear anything. They are not. Tigers are rare.

[Lecture slide omitted from the public copy.] The two distributions should not carry equal weight. Annotated in the lecture: each must be scaled by its prior, $p(\text{breeze})$ and $p(\text{tiger})$ — the probability that the stimulus was present at all, before any observation.

What changes

Weight each curve by its prior and put the threshold where the *weighted* curves cross:

$$P[+]\,p(r \mid +) \;=\; P[-]\,p(r \mid -) \qquad \Longleftrightarrow \qquad \ell(r) \;>\; \frac{P[-]}{P[+]}.$$

The test has not changed shape. Only the number on the right has moved: from 1 to the ratio of the priors. If − is nine times as common as +, you should demand nine times the evidence before calling +.

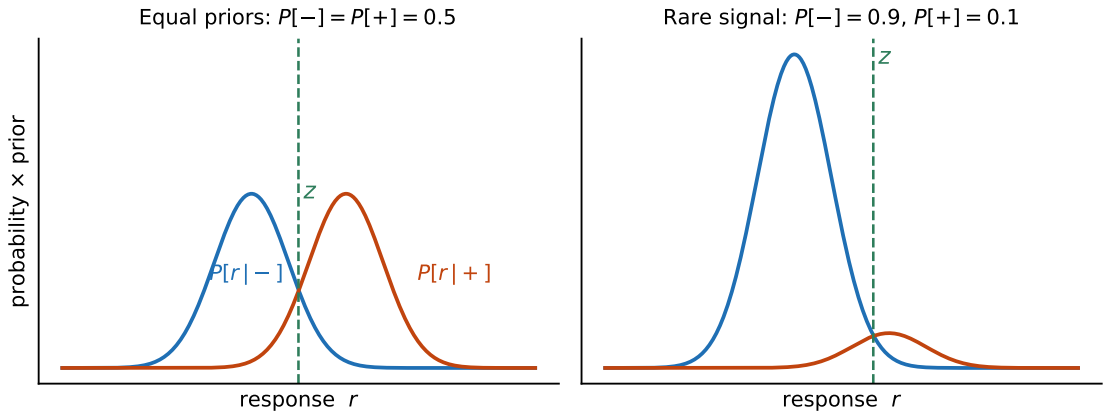


Figure 2: *Left: with equal priors the crossing point sits between the two peaks. Right: when the signal is rare the + curve shrinks, the crossing point slides right, and the threshold becomes far more conservative. The distributions themselves are unchanged — only their weights.*

Biology doing this explicitly

[Lecture slide omitted from the public copy.] Single-photon responses in a rod (Rieke lab). Left: the photoreceptor outer segments. Top: a current recording, with one photon event boxed in red and a stretch of pure noise boxed in green. Below: the resulting noise and signal distributions, already weighted by their priors.

Rods in the retina can respond to a single photon. But the noise is large relative to the signal, so a downstream cell reading the rod faces precisely our two-distribution problem: a noise distribution, and a photon-response distribution shifted along but overlapping it.

Plain signal detection theory says put the threshold at the crossing point. The synapse from photoreceptor to bipolar cell instead sets it *far* to the right, seemingly throwing away many real photon events.

[Lecture slide omitted from the public copy.] The same two distributions with the priors made explicit. At low light a genuine photon is rare, so the noise curve towers over the signal curve, the crossing point moves far to the right, and the synapse's steep threshold (the black rising curve) sits about where it should.

The idea in plain words

The synapse is not being clumsy; it has the prior built into it. At low light levels, almost every wobble is noise, so a wobble is poor evidence of a photon. Being sceptical is the correct policy, and evolution has wired that scepticism into the shape of the synaptic transfer function.

8. Costs: Not All Mistakes Are Equal

The prior fixes how *often* each situation arises. It says nothing about how much each mistake *hurts*. Mistaking a tiger for the breeze gets you eaten; mistaking the breeze for a tiger costs you a wasted sprint.

Loss

L_+ is the penalty for answering $+$ when the truth was $-$.
L_- is the penalty for answering $-$ when the truth was $+$.

Cutting your losses

For a given response r, the expected cost of each answer is its penalty times the probability that it is the wrong one:

$$\text{Loss}_+ = L_+ \, P[- \mid r], \qquad \text{Loss}_- = L_- \, P[+ \mid r].$$

Answer $+$ when doing so costs less on average: $L_+ P[- \mid r] < L_- P[+ \mid r]$.

Watch out — these are posteriors, not likelihoods

$P[- \mid r]$ is the probability that the stimulus was $-$ *given the response you just saw.* That is the reverse conditioning from $p(r \mid -)$, and the reason Bayes' rule is needed to finish the derivation.

Finishing with Bayes' rule

Bayes' rule converts each posterior into a likelihood times a prior:

$$P[- \mid r] = \frac{p(r \mid -)\, P[-]}{p(r)}, \qquad P[+ \mid r] = \frac{p(r \mid +)\, P[+]}{p(r)}.$$

Substituting into the inequality:

$$\frac{L_+ \, p(r \mid -)\, P[-]}{p(r)} \; < \; \frac{L_- \, p(r \mid +)\, P[+]}{p(r)}.$$

The denominator $p(r)$ is the same on both sides and cancels — which is the whole reason this is tractable, since $p(r)$ is the awkward term nobody wants to compute. Rearranging into the likelihood ratio:

$$\frac{p(r \mid +)}{p(r \mid -)} \; > \; \frac{L_+ \, P[-]}{L_- \, P[+]}$$

[Lecture slide omitted from the public copy.] The final decision rule, derived on the slide. Cut your losses: answer $+$ when $\text{Loss}_+ < \text{Loss}_-$. The handwritten lines substitute Bayes' rule into both sides, cancel $p(r)$, and arrive at the boxed criterion above.

The idea in plain words

The structure has not moved once in this whole lecture. It has always been

$$\text{likelihood ratio} \; > \; \text{some threshold},$$

and each section has only revised the threshold:

What you know	Threshold
Nothing beyond this trial	1
How often each stimulus occurs	$P[-]/P[+]$
. . . and what each mistake costs	$L_+P[-]\,/\,L_-P[+]$

Rarity and cost push in the same way and combine by multiplication. A rare event needs more evidence; a cheap mistake needs less. A rare event whose misses are catastrophic — the tiger — gets one factor pushing the threshold up and another pulling it down, and biology has to weigh them.

9. Where This Leaves Us

- Decoding two alternatives from a noisy response is a threshold problem, and the overlap of the two response distributions sets the floor on how well anyone can do.
- The **likelihood ratio** is the optimal statistic — by the Neyman–Pearson lemma, nothing beats it for a given false-alarm rate.
- Everything else — priors, costs — leaves the test alone and only **moves the threshold**.
- Given time, evidence accumulates as a *sum of log likelihood ratios*, wandering to a bound. LIP firing rates ramp at a coherence-dependent rate and peak at a common value before the saccade, which is what that looks like in spikes.
- A single well-chosen MT neuron matches the whole monkey's psychophysics — leaving open why cortex has so many.

> *To add from lectures: The next sections of this lecture move from one neuron to many: the **population vector**, a way of letting a large group of neurons vote on the stimulus. Fred Rieke's guest lecture covers the single-photon result in full.*

Part II

Decoding from Many Neurons: Population Codes

Lecture 2 — the population vector, Bayesian inference, maximum likelihood, and maximum a posteriori.

10. Why Many Neurons

Part I decoded a single neuron choosing between two alternatives. Two things now change at once: the stimulus becomes a *continuous* quantity rather than a binary choice, and we get to listen to a whole population rather than one cell.

The idea in plain words

A single neuron is a poor witness. Its tuning curve is broad, so many different stimuli produce much the same firing rate, and its response is noisy, so the same stimulus produces

a different rate every time. A population fixes both problems — but only if we know how to pool the votes. The rest of this lecture is three increasingly good answers to "how".

11. The Cricket Cercal System

Crickets are extremely sensitive to air movement, which is how they escape predators and fly swatters. Two antenna-like structures at the back of the abdomen, the **cerci**, are covered in fine hairs. Air motion bends the hairs, and neurons at their base transduce that bending into spikes.

[Lecture slide omitted from the public copy.] The cricket cercal system. The two cerci, about a centimetre long, are covered with fine mechanosensory hairs; the neurons innervating them are shown at the right. Jacobs et al., J. Exp. Biol. 2008.

Only four tuning curves

The elegant part: among this whole group of neurons there are only *four* types of tuning curve. Their peak responses lie in four directions at 45° to the animal's axis, and — as with the motor cortex neuron from Week 2 — the curves are approximately **cosines**.

Normalising each neuron's response by its own maximum,

$$\left(\frac{f(s)}{r_{\max}}\right)_a = \big[\cos(s - s_a)\big]_+$$

where s_a is neuron a's preferred direction and the $[\,]_+$ means *rectify*: keep the positive part, set everything negative to zero.

Watch out — the rectification is not a technicality

A cosine goes negative for half its range, but a firing rate cannot. So each neuron is silent for every wind direction more than 90° from its preferred one. Each cell reports only on its own half of the world — which is exactly why four of them are needed, as the question below works out.

Firing rate as a projection

Here is the move that makes the whole method work. Write the wind as a vector $\vec{v}$ and neuron a's preferred direction as a unit vector $\vec{c}_a$. The cosine of the angle between two vectors is their dot product divided by their lengths — and since we only care about wind *direction*, take $|\vec{v}| = 1$. Then

$$\left(\frac{f(s)}{r_{\max}}\right)_a = \big[\vec{v} \cdot \vec{c}_a\big]_+.$$

[Lecture slide omitted from the public copy.] The four cosine tuning curves and the four preferred directions, with the projection argument written in during the lecture. A neuron's normalised firing rate is the projection of the wind vector onto its own preferred direction.

The idea in plain words

A cosine tuning curve means the neuron is measuring *one component* of the stimulus — the shadow the wind vector casts along its own preferred axis. Four such neurons are four rulers held at four angles, each reporting how much of the wind lies along its own direction.

And if each neuron reports a component, recovering the whole vector should just be a matter of adding the components back up. That is the population vector.

The population vector

Give each neuron's preferred direction a weight equal to its firing rate, and add:

$$\vec{v}_{\text{pop}} = \sum_{a=1}^{4} \left(\frac{r}{r_{\max}} \right)_a \vec{c}_a .$$

[Lecture slide omitted from the public copy.] The population vector and how well it does. The error in the estimated wind direction stays around 6° across the full 360° range — small, but visibly systematic rather than random.

12. The Same Trick in Motor Cortex

Neurons in M1 are cosine-tuned to the direction of arm reaching, about a non-zero baseline rate r_0:

$$\left(\frac{\langle r \rangle - r_0}{r_{\max}} \right)_a = \vec{v} \cdot \vec{c}_a .$$

[Lecture slide omitted from the public copy.] A motor cortical neuron's cosine tuning curve for hand reaching direction. Note the baseline r_0 marked in red: the cell never falls silent, so the cosine is measured as a deviation from that resting level rather than from zero.

The population vector is built exactly as before, and for a large enough population it converges on the true movement direction:

[Lecture slide omitted from the public copy.] Population coding in M1. Each cluster of thin lines is one neuron's preferred direction weighted by its firing rate for a given reach; the heavy line is their sum. For sufficiently large N, $\langle \vec{v}_{\text{pop}} \rangle$ is parallel to the direction of arm movement.

This is not a historical curiosity. The population vector remains one of the leading methods for decoding motor cortex in real time, and it is what drives a cursor or a robotic arm in a brain–computer interface.

13. Why the Population Vector Is Not Enough

[Lecture slide omitted from the public copy.] The verdict on the method just built.

Two separate complaints, and it is worth keeping them apart:

- **Not general.** The whole construction rests on the firing rate being a *projection*, which is a fact about cosine tuning curves. A Gaussian tuning curve — the usual shape in sensory cortex — is not a projection onto anything, and the method has nothing to say about it.
- **Not optimal.** Every neuron gets a vote of the same kind. A cell with a sharp, reliable tuning curve and a cell that is broad and noisy contribute on equal terms, which is plainly wasteful. *Optimal* means making use of all the information in the stimulus/response distributions — including how trustworthy each neuron is.

The idea in plain words

The population vector never asks what the responses actually *mean*. It is a geometric recipe that happens to work when the tuning curves are cosines. What follows replaces the recipe with a question about probability: given the responses I observed, which stimulus best explains them?

14. Bayesian Inference: the Four Terms

[Lecture slide omitted from the public copy.] Bayes' law, with each term named.

$$\underbrace{p[s \mid \mathbf{r}]}_{\text{a posteriori}} = \frac{\overbrace{P[\mathbf{r} \mid s]}^{\text{likelihood}}\ \overbrace{p[s]}^{\text{prior}}}{\underbrace{p[\mathbf{r}]}_{\text{marginal}}}$$

Here $\mathbf{r}$ is now a whole *vector* of responses, one per neuron. The four terms:

- **Likelihood** $P[\mathbf{r} \mid s]$ — how probable this pattern of firing is if the stimulus were s. Already familiar from Part I.
- **Prior** $p[s]$ — how probable that stimulus was before we looked.
- **Marginal** $p[\mathbf{r}]$ — how probable this response is over all stimuli, $p[\mathbf{r}] = \int ds\, P[\mathbf{r} \mid s]\, p[s]$. It does not depend on s, so it is just a normalisation constant.
- **Posterior** $p[s \mid \mathbf{r}]$ — what we actually want: the probability of the stimulus given what we saw.

Two decoding strategies

[Lecture slide omitted from the public copy.] The two strategies, each named for the distribution it maximises.

Maximum likelihood (ML) and maximum a posteriori (MAP)

ML: report the s^* that maximises $P[\mathbf{r} \mid s]$.
MAP: report the s^* that maximises $p[s \mid \mathbf{r}]$.

The idea in plain words

ML asks: *which stimulus would most likely have produced what I just saw?* MAP asks: *which stimulus is most likely, given what I saw and what I already knew?* The two differ only through the prior — and if the prior is flat they coincide exactly. This is the same 1 versus $P[-]/P[+]$ story from Part I, now for a continuous stimulus.

15. A Worked Example: Gaussian Tuning and Poisson Firing

The rest of the lecture is one example carried all the way through. Take a population encoding a continuous parameter s — say the orientation of an edge — with Gaussian tuning curves:

$$f_a(s) \;=\; r_{\max} \exp\left(-\frac{1}{2} \left[\frac{s - s_a}{\sigma_a} \right]^2 \right)$$

so neuron a has preferred orientation s_a and tuning width σ_a.

[Lecture slide omitted from the public copy.] Gaussian tuning curves tiling the stimulus axis, with the coverage assumption circled: the tuning curves sum to a constant, $\sum_a f_a(s) = \text{const}$.

The three assumptions

1. **Good coverage.** The preferred stimuli tile the range uniformly, so $\sum_a f_a(s)$ is a constant independent of s.
2. **Independence.** Each neuron's variability is private to it.
3. **Poisson.** That variability follows a Poisson distribution.

The idea in plain words

Assumption 1 says that whatever the stimulus, the population as a whole fires at the same total rate — some neurons take over as others drop out. It looks like bookkeeping, but it is what makes the algebra collapse later: a term that does not depend on s vanishes the moment we differentiate.

[Lecture slide omitted from the public copy.] One trial. Each dot is a neuron's firing rate, plotted against that neuron's preferred orientation, for a stimulus at $s = 0$. The dashed line is the average over many trials; individual trials scatter around it, and the error bars are what Poisson firing predicts.

Writing down $P[\mathbf{r} \mid s]$

For a Poisson neuron the probability of k spikes in a window T is $P_T[k] = (rT)^k e^{-rT}/k!$. Substituting this neuron's mean rate — which is just its tuning curve value $f_a(s)$ — and its observed rate r_a:

$$P[r_a \mid s] \;=\; \frac{\big(f_a(s)T\big)^{r_a T}}{(r_a T)!}\, e^{-f_a(s)T}$$

and because the neurons are independent, the population probability is just the product:

$$P[\mathbf{r} \mid s] \;=\; \prod_{a=1}^{N} \frac{\big(f_a(s)T\big)^{r_a T}}{(r_a T)!}\, e^{-f_a(s)T}.$$

[Lecture slide omitted from the public copy.] The two assumptions doing their work: Poisson gives the single-neuron distribution, independence turns the population into a product.

Watch out — independence is the load-bearing assumption

Turning a joint distribution over N neurons into a product of N separate ones is an enormous simplification — without it the problem is intractable. It is also the assumption most

obviously false in real cortex, where neurons driven by a common input fire together. Section 19 returns to this.

16. Maximum Likelihood

Maximise the *log* likelihood, which turns the product into a sum:

$$\ln P[\mathbf{r} \mid s] \;=\; \sum_{a=1}^{N} \Big\{ r_a T \ln\big(f_a(s)T\big) \;-\; f_a(s)T \;-\; \ln\big((r_a T)!\big) \Big\}$$

Now differentiate with respect to s, and watch two of the three terms die:

- $\ln((r_a T)!)$ depends only on the *observed* rates, not on s. Gone.
- $\sum_a f_a(s)T$ is the total population rate, which assumption 1 made constant. Gone.

Only the first term survives, and $\frac{d}{ds}\ln f = f'/f$:

$$\frac{\partial}{\partial s}\ln P[\mathbf{r} \mid s] \;=\; T\sum_{a=1}^{N} r_a \frac{f_a'(s)}{f_a(s)} \;=\; 0.$$

Putting the Gaussian in

For a Gaussian tuning curve, $f_a'(s)/f_a(s) = -(s - s_a)/\sigma_a^2$ — the exponential and the prefactor cancel top and bottom, which is the whole reason this example is chosen. The condition becomes

$$\sum_{a=1}^{N} \frac{r_a\,(s - s_a)}{\sigma_a^2} \;=\; 0$$

and solving for s gives the estimate:

$$s^* \;=\; \frac{\sum_a r_a s_a/\sigma_a^2}{\sum_a r_a/\sigma_a^2}$$

[Lecture slide omitted from the public copy.] The derivation as it was worked through, ending with the estimate and its simplification.

What the answer says

The idea in plain words

This is a **weighted average of preferred stimuli**. Each neuron nominates its own preferred value s_a, its vote is weighted by how hard it is firing (r_a), and — the new part — also by $1/\sigma_a^2$, the inverse of its tuning width.
A neuron with a sharp tuning curve has small σ_a, so $1/\sigma_a^2$ is large and its vote counts heavily. A broad, vague neuron is discounted. The procedure weights each neuron's opinion by its reliability — exactly the thing the population vector could not do.

And if every neuron happens to have the same width, the σ's cancel:

$$s^* \;=\; \frac{\sum_a r_a s_a}{\sum_a r_a}$$

which is the population vector again. So the earlier method was not wrong — it was the special case of this one in which all neurons are equally reliable.

17. Maximum a Posteriori

Same procedure, one extra term. Bayes' law in logs:

$$\ln p[s \mid \mathbf{r}] \;=\; \ln P[\mathbf{r} \mid s] \;+\; \ln p[s] \;-\; \ln p[\mathbf{r}]$$

and $\ln p[\mathbf{r}]$ does not depend on s, so it disappears on differentiating. What is left is the ML condition plus a prior term:

$$\sum_{a=1}^{N} r_a \frac{f_a'(s)}{f_a(s)} \;+\; \frac{p'[s]}{p[s]} \;=\; 0.$$

Taking a Gaussian prior with mean s_{prior} and width σ_{prior}:

$$s^* \;=\; \frac{T\sum_a r_a s_a/\sigma_a^2 \;+\; s_{\text{prior}}/\sigma_{\text{prior}}^2}{T\sum_a r_a/\sigma_a^2 \;+\; 1/\sigma_{\text{prior}}^2}$$

[Lecture slide omitted from the public copy.] The MAP derivation, with the prior sketched at the right: a narrow prior pulls the estimate hard towards s_{prior}, a broad one barely moves it.

The idea in plain words

The prior has simply joined the population as one more voter. It nominates s_{prior} and votes with weight $1/\sigma_{\text{prior}}^2$ — the same inverse-variance weighting every neuron gets. A confident prior is a loud voice; a vague one is a whisper.

Two limits make this concrete. As $\sigma_{\text{prior}} \to 0$ the prior term dominates and the estimate is dragged to s_{prior} almost regardless of the data. As $\sigma_{\text{prior}} \to \infty$ the prior term vanishes and MAP becomes ML exactly.

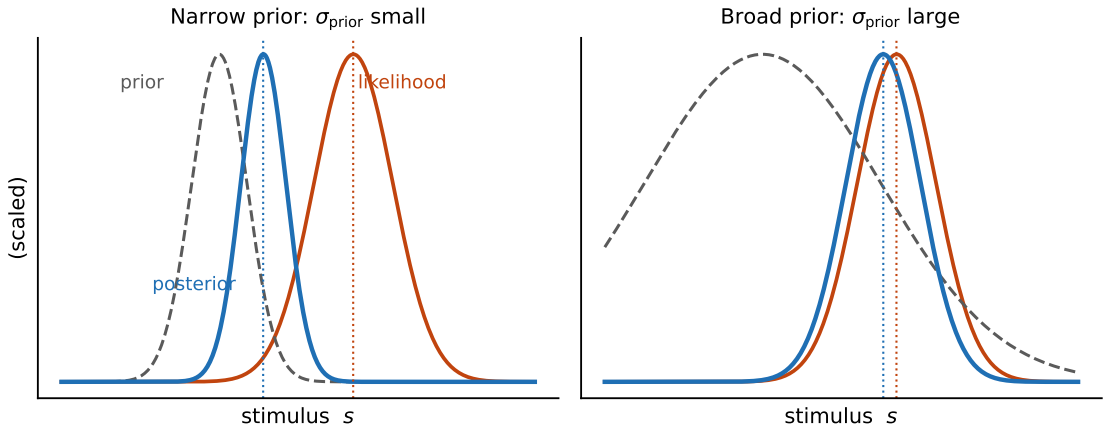


Figure 3: *The same likelihood under two priors. Left: a narrow prior centred well below the likelihood peak drags the posterior most of the way towards itself. Right: a broad prior leaves the posterior essentially where the likelihood put it. Dotted lines mark the ML estimate (rust) and the MAP estimate (blue).*

18. Limitations

[Lecture slide omitted from the public copy.] Where this framework stops.

- **It lives entirely in tuning curves and mean firing rates.** Rapid time variation in the stimulus is nowhere in the formulation — the response was a single number per neuron. Reconstructing a time-varying stimulus is the subject of the rest of this lecture.
- **It assumes independence.** Everything from $P[\mathbf{r} \mid s] = \prod_a P[r_a \mid s]$ onwards depends on it, and real cortical neurons are correlated.

[Lecture slide omitted from the public copy.] Why correlation is hard to ignore. Calcium imaging records many neurons at once, and the traces are visibly not independent. Understanding these correlations, experimentally and theoretically, is a major current research programme. Image: Alex Kwan, UC Berkeley.

The idea in plain words

If two neurons share their noise, their two votes are not two independent pieces of evidence — pooling them as though they were overstates your confidence. In the worst case, adding more correlated neurons stops adding information altogether. This is why correlation is not a detail to be tidied up later; it changes what a population code can carry.

19. Where This Leaves Us

- The **population vector** adds up preferred directions weighted by firing rate. Elegant, still used in brain–computer interfaces, but it needs cosine tuning and treats every neuron as equally trustworthy.
- **Maximum likelihood** asks which stimulus best explains the observed responses. With Gaussian tuning and Poisson firing it gives a weighted average of preferred stimuli, each neuron weighted by $1/\sigma_a^2$ — its reliability.
- When all neurons are equally reliable, that estimate reduces exactly to the population vector. The old method is a special case of the new one.
- **MAP** adds the prior as one more voter, weighted by $1/\sigma^2_{\text{prior}}$. A sharp prior biases the estimate; a flat one recovers ML.
- The framework assumes independence and ignores time. Both are load-bearing, and both are where the subject goes next.

> *To add from lectures: The remainder of this lecture covers **stimulus reconstruction** — recovering a complete time-varying stimulus from neural activity. Eric Shea-Brown gives a guest lecture on correlations in population codes.*

Part III

Reconstructing the Stimulus

Lecture 3 — the conditional mean, stimulus reconstruction, and two attempts at reading minds.

20. From a Value to a Signal

Both previous lectures decoded a *single* quantity: a binary choice in Part I, a continuous parameter in Part II. Now the target becomes a stimulus that varies continuously in time, $s(t)$, estimated from responses that also vary in time.

The idea in plain words

The motivating fantasy is recording brain activity overnight and playing back the dream in the morning. Dreaming is far out of reach, but reconstructing *waking* sensory experience from neural activity turns out to be surprisingly tractable — and it needs almost no new machinery, just a careful answer to a question we have so far dodged.

That question: ML and MAP both report the stimulus at a *peak* of a distribution. Is the peak really the best single number to report?

21. The Best Estimator Is the Conditional Mean

Ask for an estimator s_{Bayes} that is as close as possible to the true stimulus *on average*. To make "close" precise, introduce an error function $L(s, s_{\text{Bayes}})$ and minimise its average over every stimulus consistent with the observed response:

$$\frac{\partial}{\partial s_{\text{Bayes}}} \int ds\; L(s, s_{\text{Bayes}})\, p[s \mid \mathbf{r}] \;=\; 0.$$

Take the natural choice, least squares: $L = (s - s_{\text{Bayes}})^2$. The derivative brings down a factor of two and drops the square:

$$-2 \int ds\; (s - s_{\text{Bayes}})\, p[s \mid \mathbf{r}] \;=\; 0 \qquad \Longrightarrow \qquad \int ds\; s\, p[s \mid \mathbf{r}] \;=\; s_{\text{Bayes}} \int ds\; p[s \mid \mathbf{r}].$$

The remaining integral is over a normalised probability distribution, so it is just 1, and s_{Bayes} comes straight out:

$$s_{\text{Bayes}} \;=\; \int ds\; p[s \mid \mathbf{r}]\, s$$

[Lecture slide omitted from the public copy.] The derivation as worked through in the lecture, ending with the recognition written in red at the bottom right.

The conditional mean

Under squared-error cost, the best estimate of the stimulus is the *mean* of the posterior distribution — not its peak.

Watch out — this is a different answer from MAP

MAP reports where the posterior is highest; this reports where its centre of mass is. For a symmetric, single-peaked posterior they coincide. For a skewed or multi-peaked one they do not, and which is "best" depends entirely on the cost function you chose. Least squares gives the mean; a different penalty gives a different estimator.

The recognition

Now look at what that integral says when the response is a *single spike*. It takes every stimulus, weights it by the probability that it occurred given that a spike happened, and averages.

The idea in plain words

That is the **spike-triggered average**. The same quantity computed in Week 2 by grabbing the stimulus before each spike and averaging — the thing `compute_sta` calculates — turns out to be the optimal least-squares decoder for a single spike. The encoding tool and the decoding tool are the same object, arrived at from opposite directions.

22. Reconstruction by Pasting

This suggests an almost embarrassingly simple algorithm: every time a spike occurs, paste in a copy of the spike-triggered average, and add up the overlaps.

[Lecture slide omitted from the public copy.] At a low firing rate the spikes are sparse, the pasted copies barely overlap, and the reconstruction is a row of isolated bumps rather than anything resembling a smooth signal.

[Lecture slide omitted from the public copy.] At a higher firing rate the copies overlap heavily and sum into a continuously varying estimate. More spikes per unit time means a better-resolved reconstruction.

The sign problem

There is a catch, and it is the same catch as the cricket's.

Watch out — a one-signed filter cannot represent a two-signed stimulus

If the spike-triggered average has a single positive lobe, then every pasted copy pushes the reconstruction the same way. No arrangement of spikes can produce a *negative* excursion. Such a decoder can only ever report stimulus fluctuations of one sign.

This is not hypothetical — it is exactly the limitation of the fly H1 data set. The fly has *two* H1 neurons, one encoding leftward motion and one rightward. Reconstruct from a single H1 and you recover only one direction of motion; the other half of the stimulus is invisible.

23. Reading Minds I: the LGN

The first real demonstration. Recordings were made from many neurons in the **lateral geniculate nucleus** of an anaesthetised cat watching a film. Each neuron's spike train was convolved with that neuron's spatiotemporal receptive field, and the results summed.

[Lecture slide omitted from the public copy.] A frame of the reconstructed movie — noisy, low resolution, but plainly a recognisable scene. Work by Yang Dan, then a PhD student, at UC Berkeley.

The idea in plain words

This is the pasting algorithm run in parallel across a population and in two spatial dimensions rather than one temporal one. Each neuron contributes its own receptive field, stamped down at each of its spike times, and the scene emerges from the sum. No new principle —

just the conditional mean, applied at scale.

24. Reading Minds II: fMRI

Fast forward to 2011. Nishimoto and colleagues in Jack Gallant's lab recorded human visual cortex with fMRI while subjects watched films, and reconstructed one-second movie clips from the recordings.

[Lecture slide omitted from the public copy.] Presented clips on the top row, reconstructions beneath. These are stills from one-second sequences. Nishimoto et al., Current Biology 2011.

The strategy

[Lecture slide omitted from the public copy.] The whole approach in one line — the same relation used throughout Parts I and II.

The researchers look for the movie clip s that maximises the posterior $p[s \mid \mathbf{r}] \propto P[\mathbf{r} \mid s]\, p[s]$. Two ingredients:

- **The prior.** A library of 18 million clips, with $p[s]$ taken to be uniform across them. The prior is not a formula — it is a *catalogue* of what the world plausibly looks like.
- **The likelihood.** This needs a way to predict the response to an arbitrary clip, which is precisely an **encoding model** — fitted from a separate training set of movies.

The idea in plain words

This is why encoding and decoding sit in the same course. You cannot decode by inverting the brain directly. Instead you build a forward model good enough to predict responses, run it over millions of candidate stimuli, and keep the candidates whose predicted response matches what you actually measured. The decoder is an encoder run in a loop.

The lecture leaves a follow-up hanging: how would you work out the *dimensionalities* of $\mathbf{r}$ and s? It is worth answering, because the answer explains why the method has to be built the way it is.

- $\mathbf{r}$ has one component per **voxel** recorded — so its dimension is however many voxels the imaged region contains.
- s has one component per pixel, per colour channel, per frame of the clip. Even a small, short clip runs to hundreds of thousands of numbers.

The idea in plain words

So s is enormously higher-dimensional than $\mathbf{r}$. There are vastly more ways the world could have looked than there are numbers coming out of the scanner, which means the problem is hopelessly *underdetermined*: no amount of algebra inverts it, because a great many different clips are consistent with the same measured response.
That is precisely why the 18-million-clip library exists. The prior does not refine an answer that the data nearly determined — it supplies the constraint that makes an answer possible at all, by restricting the search to stimuli that actually occur in the world.

The encoding model

[Lecture slide omitted from the public copy.] The encoding model for one voxel. Top: motion-energy filters feed hemodynamic response filters to produce a predicted BOLD signal. Bottom: inside one motion-energy filter — colour conversion, space–time Gabor filtering, a quadrature pair squared and summed, a compressive log nonlinearity, and temporal downsampling from 15 Hz to 1 Hz.

Almost every stage is Week 2 material reused:

- **Space–time Gabor filtering** — the spatiotemporal linear filter.
- **Quadrature pair, squared and summed** — two filters at different phases whose squares are added. This gives a large response regardless of where in its cycle the pattern sits, which is the standard construction for **phase invariance** in a complex cell.
- **Compressive nonlinearity** — here a logarithm. The g of the linear–nonlinear model.
- **Temporal downsampling**, 15 Hz to 1 Hz, to reduce noise.
- **A separate hemodynamic filter**, because BOLD is a slow, blurred proxy for neural activity rather than the activity itself.

Watch out — the neural model and the BOLD model are fitted separately

The slow haemodynamic response is not a property of the neurons — it is a property of the measurement. Keeping it in its own filter stage means the neural part of the model can stay fast and be interpreted on its own terms.

Ranking, and why the best guess is not the best answer

[Lecture slide omitted from the public copy.] The full procedure. An encoding model is fitted per voxel and used to predict responses to every clip in the library; clips are ranked by how well their predicted response matches the measured one. Top right: the 30 highest-posterior clips. Bottom right: cumulative averages over increasing numbers of them.

The MAP solution is simply to read off the single highest-ranked clip. In practice, averaging the top 30 does considerably better.

25. Where This Leaves Us

- Under squared-error cost the optimal estimate is the **conditional mean** of the posterior, $\int ds\, p[s \mid \mathbf{r}]\, s$ — its centre of mass, not its peak.
- For a single spike that integral *is* the spike-triggered average. The Week 2 encoding tool turns out to be the optimal single-spike decoder.
- Pasting the STA at every spike reconstructs a time-varying stimulus, and works better the higher the firing rate — but a one-signed filter can only report one-signed stimuli, which is why the fly needs two H1 neurons and the cricket four cercal cells.
- Population reconstruction works: LGN spike trains convolved with receptive fields give a recognisable movie.
- Modern reconstruction inverts an **encoding model** against a large library of candidate stimuli. Averaging the best-ranked candidates beats taking the single most probable one.

*To add from lectures: A guest lecture by Fred Rieke covers retinal processing and the single-photon result from Part I in full. Week 4 moves on to **information theory** — how information is defined, what it quantifies, and what it is good for in neuroscience.*

Part IV

Guest Lecture: Seeing by Starlight

Fred Rieke — how the retina pulls single-photon signals out of an array of noisy detectors, and why it throws most of them away.

26. Vision at the Physical Limit

A long history of behavioural measurement establishes something remarkable: in very dim light, what limits our ability to see is not biological noise or inefficiency. It is the physical fact that light arrives in discrete **photons**. The visual system has been pushed up against the graininess of light itself.

[Lecture slide omitted from the public copy.] Electron micrograph of frog retina. The long cylinders are rod photoreceptor outer segments, where light is transduced into an electrical signal; the shorter, broader cells interspersed among them are cones, which take over at higher light levels.

The dark-adapted visual system can detect the absorption of just a few photons, spread across a pool of several hundred to a thousand rods. Those few absorptions are reliably detected by the retina *and* reliably transmitted onward, where they give rise to perception.

27. The Problem: Sparse Signals in Noisy Detectors

State it plainly. A handful of rods in a pool of a thousand have absorbed a photon. *Every* rod in the pool — including all the ones that absorbed nothing — is generating noise. How should the retina combine signals across that array so that the few real responses survive and the rest are rejected?

Watch out — averaging, the usual remedy for noise, is a disaster here

Averaging works when a weak signal is spread evenly across many detectors, so that summing builds the signal while the noise partly cancels. Here the signal is not spread evenly — it is concentrated in a tiny fraction of the detectors. Averaging over the whole pool buries those few real responses under the noise of the 999 rods that have nothing to contribute.

The idea in plain words

Rieke's analogy: you are in a stadium with a thousand people shouting, and you care about what a few of them are saying. Putting one microphone at the halfway line and averaging everything is hopeless. What you would do instead is go seat by seat, decide for each person whether they are likely to be one of the ones you care about — using some prior expectation, say that they are louder than average — keep those, discard the rest, and only then combine what is left.
Select first, average second. That ordering is the whole design problem.

So what is needed is a **thresholding nonlinearity**: a step that retains responses from rods likely to have caught a photon and rejects the rest. And the stakes are steep, precisely because the signal is so sparse — discarding noise from the 99.9% of rods that caught nothing, while keeping the 0.1% that did, is worth a great deal.

28. Where the Threshold Has to Live

Anatomy narrows down the possibilities sharply. Rod signals leave the retina through a dedicated circuit whose first stage is the **rod bipolar cell**, and each rod bipolar cell collects input from many rods.

[Lecture slide omitted from the public copy.] Convergence in the mammalian retina. Many rods feed a common downstream target; a couple of them (red) have absorbed photons. At the right, the measured distributions of rod response amplitude — noise, and single-photon responses — which overlap.

The idea in plain words

Once the rod bipolar cell has pooled its inputs *linearly*, weighting noisy rods and signal-carrying rods equally, the averaging disaster has already happened: signal and noise are inextricably mixed and no later stage can undo it.
So the last moment at which a per-rod decision is still possible is the synapse *between* the rod and the rod bipolar cell. If a threshold exists anywhere, it has to be there — the anatomy leaves nowhere else.

Direct recordings from rod bipolar cells bear this out: there is a thresholding nonlinearity at that synapse, just where the argument says it must be.

[Lecture slide omitted from the public copy.] The full picture. Left: the rod circuit through rod bipolar and AII amacrine cells to the ganglion cell. Right: the signal and noise distributions with the estimated synaptic nonlinearity overlaid in blue — everything below the transition is thrown away, everything above is kept. Note the twist flagged at the bottom.

29. The Puzzle

The nonlinearity is in the right place. Its *position along the amplitude axis* is the surprise.

Reading the distributions naively, you would put the threshold at the point where the noise and signal curves cross. That is the maximum-likelihood rule from Part I: keep a response if its amplitude is more likely to have come from the single-photon distribution than from the noise distribution.

Watch out — the measured threshold sits far to the right of the crossing point

It is pushed well *up into* the single-photon response distribution — which means the synapse discards a large majority of genuine single-photon responses. For a system whose whole purpose is seeing in near-darkness, that looks exactly backwards.

30. The Resolution: the Prior

The naive reading has an omission, and it is the same one from Part I: the plot takes no account of *how often* each case actually arises.

[Lecture slide omitted from the public copy.] The same two distributions plotted with their prior probabilities built in. Left: at about one absorbed photon per rod, signal and noise carry comparable weight and the crossing point sits low. Right: near visual threshold, where roughly one rod in ten thousand absorbs a photon, the signal distribution shrinks by four orders of magnitude, the crossing point moves far to the right — and lands close to the measured nonlinearity (blue).

The idea in plain words

The area under each curve is the probability of being in that situation at all. Drawn with equal areas, the two hypotheses look equally plausible before you measure anything. But near visual threshold they are nothing of the sort: noise is roughly ten thousand times more common than a real photon absorption.
Scaling the curves by that prior shrinks the signal distribution dramatically, which drags the crossing point far to the right — and that is where the synapse's threshold actually sits. Plain maximum likelihood, applied to properly weighted distributions, predicts the measured position.

31. What This Adds

[Lecture slide omitted from the public copy.] The three places the problem has to be solved: transducing single photons reliably, transmitting them reliably across the synapse, and coding them so that a few absorptions produce a detectable change in the optic nerve.

- Where signals are **sparse** across an array of noisy detectors, you must **select before you average**. Averaging first destroys exactly what you were trying to recover.
- Anatomy pins down where selection must happen: at the last point before convergence mixes signal and noise irreversibly — the rod to rod bipolar synapse.
- The position of that threshold is not a biological accident. It is what maximum likelihood predicts *once the prior is included*, and it is wrong by orders of magnitude if the prior is left out.
- Part I introduced this result as a cartoon. The full version makes the stronger claim: the retina is not merely influenced by the prior, its synaptic transfer function *is* the prior, built into the hardware.